# 3.6 Using the Quadratic Formula

Name _____ Period ____ Date _____

Algebra II 3.6 · Student sheet

**I can…**

- solve quadratic equations using the Quadratic Formula.
- use the discriminant to find the number and type of solutions.
- use the Quadratic Formula to solve real-life problems.

**Vocabulary** · fill in the missing word

**Quadratic Formula** — $x =$ _____, for $ax^2 + bx + c = 0$

**Discriminant** — _____

**$b^2 - 4ac > 0$** — two real _____

**$b^2 - 4ac = 0$** — one real _____

**$b^2 - 4ac < 0$** — two _____ solutions

## Part 1 · Discriminant sort

| Equation | $b^2 - 4ac$ | Predicted x-intercepts | Graph agrees? |
|---|---|---|---|
| $x^2 - 4x + 4 = 0$ | | | |
| $x^2 + 2x + 5 = 0$ | | | |
| $x^2 - x - 6 = 0$ | | | |
| $2x^2 + 3x + 1 = 0$ | | | |
| $x^2 + 6x + 9 = 0$ | | | |
| $3x^2 - x + 2 = 0$ | | | |

## Part 2 · Notes and examples (fill in each note, then work the example)

**Quadratic Formula:** $x =$ _____

### Example 1

Solve $x^2 + 3x - 5 = 0$ using the Quadratic Formula.

### Example 2

Solve $2x^2 - 4x + 5 = 0$.

**The discriminant:** Discriminant _____ > 0: _____; = 0: _____; < 0: _____.

### Example 3

Find the discriminant and describe the solutions.

a. $4x^2 - 12x + 9 = 0$ b. $x^2 + 2x + 7 = 0$ c. $3x^2 + x - 2 = 0$

### Example 4

A ball is thrown upward. Its height h (in feet) after t seconds is $h = -16t^2 + 30t + 6$. When does it hit the ground?

## Part 3 · Partner practice

### Partner 1

Solve $x^2 - 6x + 4 = 0$.

### Partner 2

Find the number and type of solutions of $5x^2 + 2x + 1 = 0$.